Environment Setup

Contents

1 Purpose

This document explains how to install Python, GPU libraries, and Jupyter on your machine so you can run the lab notebooks. The course package contains `labs/` (notebooks), `notes/` (PDF modules), `datasets/` (data files), `requirements.txt`, `setup.bat`, and `env.ps1`.

2 What You Need

2.1 Hardware

- Windows 10 or 11 (64-bit).
- NVIDIA GPU with a recent driver (recommended). A 4 GB card works if you use the batch sizes given in each module.
- 16 GB RAM minimum (32 GB if you run Spark and Jupyter together).
- About 15 GB free disk space for the virtual environment, PyTorch, and cached model weights.

2.2 Install Before You Start

1. **Python 3.10 or 3.11** from `https://www.python.org/downloads/`. Enable "Add Python to PATH" during installation.
2. **NVIDIA driver** supporting CUDA 12.x (driver version 525 or newer is a safe baseline for CUDA 12.1 wheels).
3. **Microsoft OpenJDK 17** (only required for the Spark lab in Module 18): `https://learn.microsoft.com/en-us/java/openjdk/download`.

3 First-Time Installation

Open PowerShell in the course folder (the directory that contains `setup.bat`).

3.1 Option A: use setup.bat

```
setup.bat
```

This creates a virtual environment, installs PyTorch with CUDA 12.1, installs packages from `requirements.txt`, configures Spark support for Windows, registers a Jupyter kernel, and downloads datasets into `datasets/`. Wait until it finishes without errors.

3.2 Option B: manual install

If `setup.bat` fails, run these commands one block at a time:

```
python -m venv .venv
.venv/Scripts/activate
python -m pip install --upgrade pip setuptools wheel
pip install torch torchvision torchaudio ^
  --index-url https://download.pytorch.org/whl/cu121
pip install -r requirements.txt
python -m ipykernel install --user --name lab-env ^
  --display-name "Python 3.11 (labs)"
```

For Module 18 (Spark), install JDK 17 and ensure `tools/hadoop/bin/winutils.exe` exists in the course folder. After installing JDK, set `JAVA_HOME` to the JDK path (see Section 4).

4 Every Session

Each time you open a new terminal:

```
cd C:/path/to/course-folder
.venv/Scripts/activate
. ./env.ps1
```

`env.ps1` sets cache paths for Hugging Face models, Spark/Hadoop variables for the Spark lab, and MLflow tracking for Lab 19. Run it after activating the venv.

Confirm the GPU is visible:

```
python -c "import torch; print(torch.cuda.is_available())"
python -c "import torch; print(torch.cuda.get_device_name(0) if torch.cuda.is_available() else
↪  'CPU only')"
```

If this prints `False`, PyTorch was installed without CUDA. Reinstall:

```
pip install torch torchvision torchaudio ^
  --index-url https://download.pytorch.org/whl/cu121
```

5 Running Labs

Start Jupyter:

```
jupyter lab
```

Select kernel **Python 3.11 (labs)** (or the kernel registered during setup). Open notebooks from `labs/`. Read the matching PDF in `notes/` first.

6 Datasets

The `datasets/` folder should contain MNIST, sample images, a credit-fraud parquet file, instruction JSONL for LoRA, and YOLO weights. If a lab reports missing data, re-run `setup.bat` or ask for a fresh dataset bundle. Lab 09 downloads SST-2 sentiment data from Hugging Face on first run (requires internet and `env.ps1` loaded).

7 Spark Lab (Module 18 Only)

Spark needs JDK 17 and `HADOOP_HOME` pointing at `tools/hadoop` in the course folder. `env.ps1` sets these when JDK is installed in the default Microsoft path. If Spark fails with a `winutils` or `UnsatisfiedLinkError` message, confirm `tools/hadoop/bin/winutils.exe` exists and restart the terminal after sourcing `env.ps1`.

8 MLflow (Lab 19)

With `env.ps1` loaded:

```
mlflow ui --backend-store-uri ./mlruns
```

Open `http://127.0.0.1:5000` in a browser to compare logged runs.

Table 1: Module, study PDF, and lab notebook.

Mod.	PDF	Notebook
00	`00-environment-setup.pdf`	(this guide)
01	`01-rnns-sequence-modeling.pdf`	`lab01_rnn_sequence_modeling.ipynb`
02	`02-lstm-gru.pdf`	`lab02_lstm_gru_comparison.ipynb`
03	`03-attention-seq2seq.pdf`	`lab03_attention_seq2seq.ipynb`
04	`04-transformers.pdf`	`lab04_transformer_block.ipynb`
05	`05-gan-fundamentals.pdf`	`lab05_gan_mnist.ipynb`
06	`06-dcgan-gan-training.pdf`	`lab06_dcgan_mnist.ipynb`
07	`07-diffusion-theory.pdf`	`lab07_diffusion_visualization.ipynb`
08	`08-ddpm-stable-diffusion.pdf`	`lab08_ddpm_mnist.ipynb`
09	`09-bert.pdf`	`lab09_bert_finetune.ipynb`
10	`10-gpt-causal-lms.pdf`	`lab10_gpt_generation.ipynb`
11	`11-lora-peft.pdf`	`lab11_lora_finetune.ipynb`
12	`12-faster-rcnn.pdf`	`lab12_faster_rcnn_inference.ipynb`
13	`13-yolo-map.pdf`	`lab13_yolo_training.ipynb`
14	`14-segmentation-unet-maskrcnn.pdf`	`lab14_segmentation.ipynb`
15	`15-rl-mdps-qlearning.pdf`	`lab15_qlearning_frozenlake.ipynb`
16	`16-dqn.pdf`	`lab16_dqn_cartpole.ipynb`
17	`17-policy-gradients-actor-critic.pdf`	`lab17_a2c_cartpole.ipynb`
18	`18-spark-mllib.pdf`	`lab18_spark_mllib.ipynb`
19	`19-ddp-mlops.pdf`	`lab19_ddp_mlflow.ipynb`
20	`20-shap-lime-xai-deployment.pdf`	`lab20_xai_deployment.ipynb`

9 Troubleshooting

9.1 CUDA out of memory

Lower `batch_size` in the notebook. Modules 09 to 11 list safe values for 4 GB GPUs. Close browser GPU tabs and other CUDA applications.

9.2 Missing Python package

Activate `.venv`, then: `pip install -r requirements.txt`.

9.3 Wrong Jupyter kernel

Kernel menu → Change kernel → **Python 3.11 (labs)**. If missing, run:

```
python -m ipykernel install --user --name lab-env ^
  --display-name "Python 3.11 (labs)"
```

9.4 Hugging Face download fails

Run `env.ps1` first. Check internet access. Lab 09 uses dataset `SetFit/sst2` (not GLUE SST-2).

9.5 Package list

`requirements.txt` includes PyTorch, transformers, diffusers, ultralytics, gymnasium, pyspark, mlflow, shap, lime, jupyter, and standard scientific Python libraries. Do not upgrade packages mid-course unless a lab explicitly requires it.